Experiment 5: Analysis of Data Communication in a Web Application Using Browser Developer Tools

Aim

To analyze the data communication between a browser and a web server using the browser's developer tools.

Objective

- Understand the HTTP request-response cycle.
- Use the Network tab to inspect requests, headers and status codes.
- Observe loading time and data size of web resources.

Tools Required

Google Chrome (or Microsoft Edge / Firefox) and any public website or a sample login page.

Theory

When a user opens a web page, the browser sends an **HTTP request** to the server (method, URL, headers). The server returns an **HTTP response** (status code, headers, body). A single page usually needs many requests for HTML, CSS, JavaScript, images and API data.

Status Code	Meaning
200	OK - request successful
301 / 302	Redirect
304	Not modified - cached copy used
404	Resource not found
500	Internal server error

Procedure

1. Open Chrome and press **F12** (or right-click > Inspect) to open Developer Tools.
2. Click the **Network** tab and tick 'Disable cache' for a fresh load.
3. Enter the website address and press Enter; reload with Ctrl + R.
4. Observe the list of requests: name, status, type, size and time.
5. Click the first request (the HTML document) and study the **Headers**, **Preview**, **Response** and **Timing** tabs.
6. Use the filter buttons (Doc, CSS, JS, Img, Fetch/XHR) to group the resources.
7. Log in to a sample form and watch the POST request and its payload under the Payload tab.
8. Note the total requests, data transferred and load time shown at the bottom of the panel.

Attach screenshot here - Network tab with request list

Attach screenshot here - Headers tab of one request

Sample Observation Table

Replace these sample values with what you actually see on your screen.

Request	Method	Status	Type	Size	Time
index.html	GET	200	document	18 KB	120 ms
style.css	GET	200	stylesheet	9 KB	60 ms
app.js	GET	200	script	75 KB	140 ms
logo.png	GET	304	png	cached	15 ms
/api/login	POST	200	fetch	1 KB	210 ms

Key Points Observed

- The request headers include Host, User-Agent, Accept and Cookie.
- The response headers include Content-Type, Content-Length, Cache-Control and Set-Cookie.
- A secure site uses HTTPS, so data is encrypted in transit.
- Login data is sent in the request body with the POST method, not in the URL.
- Cached files return status 304 and load faster.

Results

The request-response cycle of a web application was analyzed, and the headers, status codes, payload and loading time of different resources were identified.

Conclusion

Browser developer tools help developers debug failed requests, check API responses and improve page speed. Understanding data communication is important for building fast and secure web applications.